

JACOPO GHIRRI

Milan, Italy

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Summary

R&D engineer and AI researcher with a PhD in mathematical modeling, deep learning and statistical inference. 3+ years of research experience developing machine learning models (generative models, deep time-series forecasting, variational inference) for large-scale policy and environmental data. Currently working in ML/AI research and development.

Experience

- Kline** **Jul. 2026 – Present**
ML/AI Research and Development *Milan, Italy*
- Apply machine learning and AI to risk management, forecasting and optimization problems in the financial sector, as part of the firm's applied R&D activities.
 - Translate quantitative research into methodologies for fintech consulting clients, drawing on deep learning, statistical inference and Python-based modeling.
- Mila – Quebec AI Institute** **Sep. 2025 – Feb. 2026**
Visiting Researcher *Montreal, Canada*
- Contributed to a Python/PyTorch repository on generative flow networks (GFlowNets) and adapted it to explore high-dimensional policy spaces in macroeconomic climate simulators.
 - Collaborated with Prof. Alex Hernandez-Garcia on bridging Integrated Assessment Models and deep generative AI architectures.
- RFF-CMCC European Institute on Economics and the Environment (EIEE)** **Jun. 2023 – Jun. 2026**
Affiliated Researcher *Milan, Italy*
- Built a deep learning forecasting pipeline in Python for short-term EU CO₂ emission prediction across 27 countries and multiple sectors.
 - Developed a robust decision-making framework under model uncertainty using multiplier preferences and statistical divergence measures.
 - Built function-on-scalar regression emulators of integrated assessment models, with Gaussian-process residual sampling and Monte Carlo propagation of climate and damage uncertainty.
- Accenture** **Jan. 2023 – May 2023**
Enterprise Architect Analyst *Milan, Italy*
- Conducted enterprise architecture assessments and data-flow mapping for large-scale IT transformation projects.

Education

- Politecnico di Milano** **Jun. 2023 – Sep. 2026**
Ph.D., Science, Technology and Policy for Sustainable Change *Milan, Italy*
- Research focus: data science, machine learning and AI for robust decision-making and uncertainty quantification (deep learning, generative models, Bayesian and variational methods).
 - Thesis: *Evaluating Uncertainties and Robust Strategies in Climate Mitigation Scenarios*. Supervisor: Prof. Massimo Tavoni. Horizon Europe projects: EUNICE (ERC), GEOCEP.
- Politecnico di Milano** **Sep. 2020 – Dec. 2022**
M.Sc. in Mathematical Engineering (Statistical Learning), cum laude *Milan, Italy*
- Thesis: *An evaluation of researchers' migrations in Europe using digital trace data*.
- Politecnico di Milano** **Sep. 2017 – Jul. 2020**
B.Sc. in Mathematical Engineering *Milan, Italy*

Publications

Decision-making under model misspecification: an application to climate policy
J. Ghirri, M. Marinacci, L. Berger, M. Tavoni. *Decisions in Economics and Finance*, 2026. [doi:10.1007/s10203-026-00579-4](https://doi.org/10.1007/s10203-026-00579-4)

Europe's Climate Ambition Under Scrutiny: Evidence from Deep Learning Emission Projections
J. Ghirri et al. *arXiv preprint*, 2026. [arXiv:2608.18690](https://arxiv.org/abs/2608.18690)

Variational inference for semiparametric Bayesian novelty detection in large datasets
L. Benedetti, E. Boniardi, L. Chiani, J. Ghirri, M. Mastropietro, A. Cappozzo, F. Denti. *Advances in Data Analysis and Classification*, 18, 681–703, 2024. [doi:10.1007/s11634-023-00569-z](https://doi.org/10.1007/s11634-023-00569-z)

Technical Skills

Core Programming: Python (PyTorch, Scikit-learn, Pandas, NumPy), R

Machine Learning & AI: Deep learning (CNNs, LSTMs, Transformers, GFlowNets), Bayesian inference, variational methods, time-series forecasting, anomaly detection

Data & Methods: Large-scale panel data processing, feature engineering, statistical modeling, causal inference, sensitivity analysis, data visualization

Infrastructure & Tools: Cloud computing (AWS), remote HPC workflows, Linux, Git, LaTeX

Familiar with: C++, TensorFlow

Languages: Italian (native), English (professional, C1)